Best Practices for Biomedical Research Data Management

Module 7: Data Sharing and Reuse
Open Access

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This video will provide an overview of open access: what the open access movement is, what the benefits of open access can be, and a few mechanisms for publishing open access. This video will also look at the economics of the scholarly communication system in order to understand why open access is difficult to achieve.

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What is open access?

Open access (OA) is
free, unrestricted, online access
to research and scholarly products.

"OA is compatible with copyright, peer review, revenue (even profit), print, preservation, prestige, quality, career-advancement, indexing, and other features and supportive services associated with conventional scholarly literature."

-- Peter Suber (2004, updated 2015). Open Access Overview.
http://legacy.earlham.edu/~peters/fos/overview.htm

Why open access?

"I truly believe that published scientific articles must be open access, allowing this treasure to best benefit research and society."

-- Hong Yu, PhD
Professor, Dept. of Quantitative Health Sciences, Health Informatics and Implementation Science, UMass Medical School

Why open access?

Increased academic impact

Total number of studies so far	70
Studies that found a citation advantage	46
Studies that found no citation advantage	17
Studies that were inconclusive, found non-significant data or measured other things than citation advantage for articles	7

SPARC Europe (2015). The Open Access Citation Advantage Service. http://sparceurope.org/oaca/

Tennant JP, Waldner F, Jacques DC *et al.* The academic, economic and societal impacts of Open Access: an evidence-based review [version 2; referees: 3 approved, 2 approved with reservations]. F1000Research 2016, 5:632. (doi: 10.12688/f1000research.8460.2)

Scholarly communication landscape

Journal prices continue to rise

Resources remain flat

Publishers profit (5 publishers publish 52% of science, technology and medicine titles)

Perpetuated by academic reward system

!

Profit	Company	Industry
10%	BMW	automobiles
23%	Rio Tinto	mining
25%	Google	search
29%	Apple	premium computing
35%	**Springer**	**scholarly publishing**
37%	**Elsevier**	**scholarly publishing**

http://wp.me/ph4jF-km CC-BY Alex Holcombe

Alex Holcombe (May 21, 2015). Scholarly publisher profit update.
https://alexholcombe.wordpress.com/2015/05/21/scholarly-publisher-profit-update/
Library Journal Periodicals Price Survey (2016).
http://lj.libraryjournal.com/2016/04/publishing/fracking-the-ecosystem-periodicals-price-survey-2016/
Lariviere et al. (2015). The Oligopoly of Academic Publishers in the Digital Era. http://dx.doi.org/10.1371/journal.pone.0127502

Why open access? Summing up

Altruistic

Better visibility, improved dissemination, and higher impact for scholarship

More knowledge leads to better patient outcomes

Return on the public's investment in taxpayer-funded research

To help achieve science's full potential by removing price barriers

Improved education

Practical

Expectations around access have changed

Mechanisms for communication and dissemination have evolved

Article-level metrics have emerged

Journal pricing and academic reward systems have remained largely unchanged

The scholarly publishing system is out of balance

Achieving Open Access

Open access publishing (gold OA)

The publication of a scholarly article in open access in a journal, peer-reviewed and often financed through article publication charges.

Nucleic Acids Research

BioMed Central
The Open Access Publisher

PLOS

Open access publishing (gold OA)

How it works

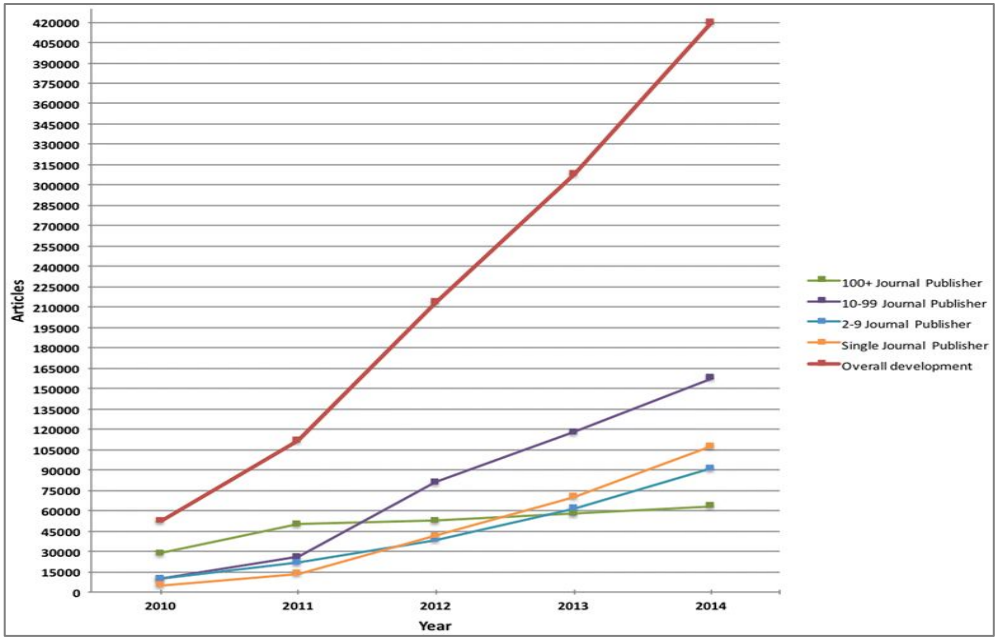
1. Select journal to submit your article: open access journal (free or paid), or subscription journal with open access option
2. Pay Article Processing Charge (if required) for immediate open publication
3. Select license (if applicable) to allow broadest dissemination and reuse possible

Harvard Open Access Project (2015). How to make your own work open access. http://cyber.law.harvard.edu/hoap/How_to_make_your_own_work_open_access

Directory of Open Access Journals (DOAJ). Available from: https://doaj.org/

“Predatory” publishers

Opportunistic open access publishers that charge publication fees to authors without providing the editorial and publishing services associated with legitimate journals.



Shen and Bjork (2015). ‘Predatory’ open access: a longitudinal study of article volumes and market characteristics. http://dx.doi.org/10.1186/s12916-015-0469-2

Evaluating open access journals

Do your homework!
Choose the right journal for your research.
http://thinkchecksubmit.org/

- ❏ Have you heard of the journal before?
- ❏ Have you read any articles in the journal?
- ❏ Have any of your colleagues published articles in the journal?
- ❏ Is the journal peer-reviewed?
- ❏ Is the journal's standard fee schedule publicly accessible?
- ❏ Do you recognize the members of the Editorial Board?
- ❏ Is the journal listed in the Directory of Open Access Journals (DOAJ)?
- ❏ Does the journal have policies and practices consistent with the Open Access Scholarly Publishers Association Code of Conduct and Principles of Transparency and Best Practice in Scholarly Publishing?
- ❏ Is the journal indexed in an established and reputable database such as PubMed, Web of Science, or Scopus?
- ❏ Does the journal have a true impact factor found in Journal Citation Reports?

Achieving Open Access

The self-archiving option (green OA)

The archiving of a scholarly publication for public access in a repository other than that of the publisher, e.g. an institutional repository or a discipline-related repository.

Cornell University Library
arXiv.org

PubMed Central

UMASS MEDICAL SCHOOL
eScholarship@UMMS

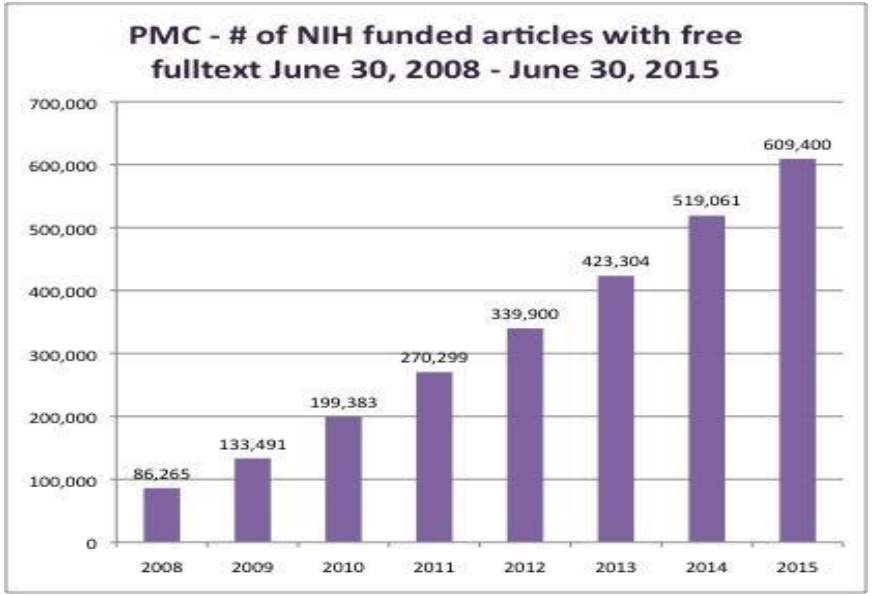
Self-archiving

How it works

1. Find out the status of your works' copyright and publisher policies for archiving (start with the SHERPA/RoMEO website)
2. Identify an appropriate open access repository
3. Deposit your work (or have someone deposit it for you)

Harvard Open Access Project (2015). How to make your own work open access. http://cyber.law.harvard.edu/hoap/How_to_make_your_own_work_open_access
Carly Strasser (2012). Researchers! Make your previous work OA. http://datapub.cdlib.org/2012/11/06/researchers-make-your-previous-work-oa/

Slow but steady growth



Growth in self-archiving is driven by funder mandates and institutional policies

Heather Morrison (June 2015). Dramatic Growth of Open Access June 30, 2015. from: http://poeticeconomics.blogspot.com/2015/06/dramatic-growth-of-open-access-june-30.html
Registry of Open Access Repository Mandates and Policies (ROARMAP). http://roarmap.eprints.org/

What you can do

Know your author rights

- Read your copyright transfer or license agreements before you sign!
- Review http://www.sparc.arl.org/initiatives/author-rights/, from the Scholarly Publishing & Academic Resources Coalition (SPARC)

Retain your right to post open access versions of your articles in open access repositories

- Scholar's Copyright Addendum Engine is easy to use
- Copyright is an author's right, not a publisher's right

Publish open access

- Publish in OA journals and/or self-archive your work
- Utilize Directory of Open Access Journals, http://www/doaj.org